

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 9- Limits and Derivatives

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1. For each of the following functions, determine whether the limit exists or does not exist at a given value of x .

(a) $f(x) := \begin{cases} 10 & \text{if } x \geq 0, \\ -10 & \text{if } x < 0, \end{cases} \text{ at } x = 0$

(c) $f(x) := \frac{x^2 - x - 2}{x - 2}, \quad x \neq 2, \text{ at } x = 2$

(b) $f(x) := \begin{cases} 7x - 2 & \text{if } x \geq 2, \\ 3x + 5 & \text{if } x < 2, \end{cases} \text{ at } x = 2$

(d) $f(x) := \begin{cases} x^2 & \text{if } x < -2, \\ -3x & \text{if } x \geq -2, \end{cases} \text{ at } x = -2$

(e) $f(x) = \frac{|x|}{x} \text{ at } x = 0$

2. Evaluate each of the following:

(a)

$$\lim_{x \rightarrow 3} \frac{\left(\frac{1}{x} - \frac{1}{3}\right)}{x - 3}.$$

(d)

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

(g)

$$\lim_{x \rightarrow 1} \frac{x^4 + 3x^3 - 13x^2 - 27x + 36}{x^2 + 3x - 4}$$

(b)

$$\lim_{x \rightarrow 3} \frac{2x}{x - 3}$$

(e)

$$\lim_{x \rightarrow 2} \sqrt{\frac{x^2 - 3x + 2}{x^2 - 4}}$$

(h)

(c)

$$\lim_{x \rightarrow -3} \frac{x^2 - 3x}{x^2 - 9}$$

(f)

$$\lim_{x \rightarrow 1} \frac{x^3 + x^2 - x - 1}{x - 1}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x + 9} - 3}{x^2}$$

3. Evaluate each of the following:

(a)

$$\lim_{x \rightarrow +\infty} 2x^{11} - 5x^6 + 3x^2 + 1$$

(d)

$$\lim_{x \rightarrow +\infty} \frac{3x^3 - 4x + 2}{7x^3 + 5}$$

(f)

$$\lim_{x \rightarrow +\infty} \frac{4x - 1}{\sqrt{x^2 + 2}}.$$

(b)

$$\lim_{x \rightarrow -\infty} 3x^4 - x^2 + x - 7$$

(e)

(g)

(c)

$$\lim_{x \rightarrow +\infty} \frac{2x + 5}{x^2 - 7x + 3}$$

$$\lim_{x \rightarrow +\infty} \frac{4x^5 - 1}{3x^3 + 7}$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 5}}{3x^2 - 2}.$$

4. Discuss the continuity of each of the following functions at a given value of x .

$$(a) f(x) := \begin{cases} x^2 & \text{if } x \leq 0, \\ x & \text{if } x > 0, \end{cases} \quad \text{at } x = 0$$

$$(c) f(x) := \begin{cases} \frac{x^2-4}{x+2} & \text{if } x \neq -2 \\ \text{at } x = -2 \end{cases}$$

$$(b) f(x) := \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x < 0, \end{cases} \quad \text{at } x = 0$$

$$(d) f(x) := \begin{cases} x+1 & \text{if } x \geq 2, \\ 2x-1 & \text{if } 1 < x < 2, \\ x-1 & \text{if } x \leq 1, \end{cases} \quad \text{at } x = 1 \text{ and } x = 2$$

5. Find the derivative of each of the following from the first principles:

$$(a) y = 1$$

$$(c) y = 2x^2 + 3x + 6$$

$$(e) y = \sqrt{x-1}$$

$$(b) y = 3x$$

$$(d) y = \frac{1}{x^2+2}$$

$$(f) y = \frac{1}{\sqrt{2x+1}}$$

6. Differentiate each of the following with respect to x :

$$(a) y = x + \frac{1}{x}$$

$$(d) y = \frac{3x^7+5x^5-2x^4+x-3}{x^4}$$

$$(b) y = x^2 - \frac{8}{\sqrt{x}}$$

$$(c) y = (x+2)^2$$

$$(e) y = 2x^3 + \sqrt{x} + \frac{x^2+2x}{x^2}$$

7. Find the derivative of each of the following:

$$(a) f(x) = \sqrt{4 - \sqrt{4+x}}$$

$$(c) f(x) = x^{\frac{1}{5}}(x-1)^{\frac{3}{5}}$$

$$(b) f(x) = \frac{1}{(3x^2+5)^4}$$

$$(d) f(x) = \frac{x}{\sqrt{x^2+1}}$$

8. Show that $f'(x) = \frac{3}{2}x^{-\frac{1}{2}}(4-x)$, if $f(x) = 12x^{\frac{1}{2}} - x^{\frac{3}{2}}$

9. Expand the right hand side of $f(x) = (x^{\frac{3}{2}} - 1)(x^{-\frac{1}{2}} + 1)$, and hence find $f'(x)$.

10. Find the value of the following derivatives at a given point.

$$(a) x^2 - xy + y^2 = 3, \frac{dy}{dx} \text{ at } (1, 2)$$

$$(d) x^3y = 2, \frac{d^2y}{dx^2} \text{ at } x = -5$$

$$(b) 2x + 3y^2 + 3x^2y + 12 = 4 \times 2^x \text{ at } (2, 0)$$

$$(e) x^2 + y^2 = 25 \text{ at } (2, 3)$$

$$(c) y = \sqrt{4-x}, \frac{dy}{dx} \text{ at } x = 5$$

$$(f) x^3 + y^3 = 6xy \text{ at } (1, 1)$$

11. Differentiate with respect to x :

$$(a) f(x) = (3x^4 - x^2)^5$$

$$(e) f(x) = 4^{3x}$$

$$(i) f(x) = \sin(x^2 + 2)$$

$$(b) f(x) = \sqrt{x-2}$$

$$(f) f(x) = \ln(2x^2 + 1)$$

$$(j) f(x) = \tan(2x + 1)$$

$$(c) f(x) = x^2\sqrt{x-2}$$

$$(g) f(x) = \log_2(5x + 1)$$

$$(d) f(x) = e^{3x^2}$$

$$(h) f(x) = \cos(2x)$$